Technical Interview Task

Human Validation Interface Generation and Implementation

Context

AI automation is becoming a standard part of daily operations. In practice, these workflows usually can be divided into three distinct sequential stages:

1. **Flow triggering logic**: what initiates the automation
2. **Agentic task execution**: one or more AI agents execute one or more tasks
3. **Human validation**: a human reviews and validates the execution result

The first two stages drive speed and scale; the third stage ensures quality, safety, and business confidence. This assignment focuses on that final stage: **human validation**. A natural objective is therefore to define a standardized, automated process for generating human validation interfaces, enabling seamless creation of automation flows that follow this 3-stage setup.

Assignment

Design and implement a practical solution that **automatically generates a human validation interface** from an automation setup involving one or more AI agents.

- The goal is **not** to build a UI for one specific run result. The goal is to automate the generation of the validation UI from the **agent(s) definition(s) themselve** (prompting structure, expected inputs, available tools, etc.).
- Your solution should include both a **technical design** and a **working implementation or prototype**.
- The implementation can be minimal, but it should demonstrate the core generation logic end to end.
- The approach should remain reusable across different automation domains

Example Scenario

The following example demonstrates a concrete scenario for your implementation.

Example Input

Agent System Prompt

```
you are an agent for the customer support of SuperNiceCompany Corporation.

your tasks are the following :
- generating responses to customer written requests based on generic
↪  instructions and historical data about elder human written
↪  question/response.
- if the customer written request can be resolved by any of you available
↪  tool, resolve it.
```

User Prompt Placeholder

```
Generic instructions :

{{generic_intructions}}

Customer mail :

{{custome_mail}}

similar questions & responses :

{{similar_questions_responses}}

answer :
```

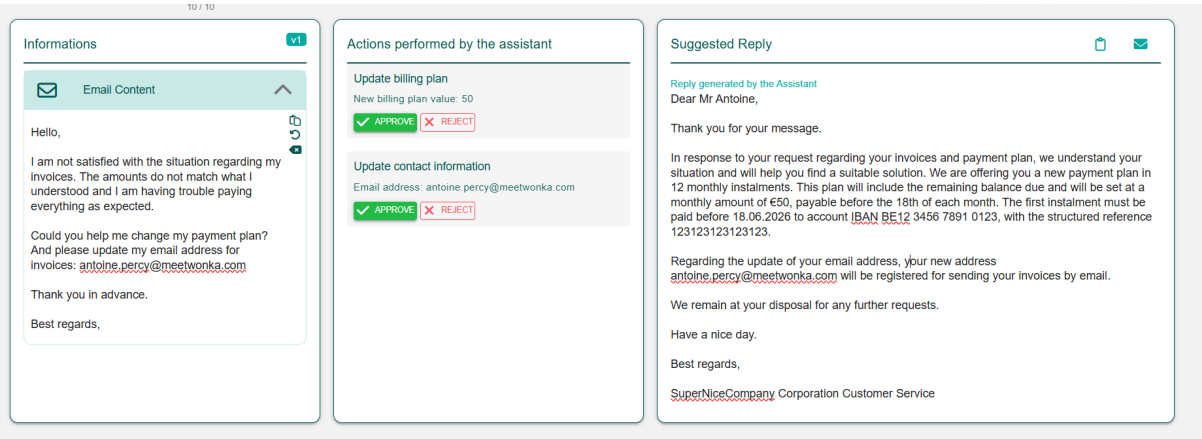
Available Tools

1. `update_customer_budget_billing_plan(new_bbp : float)`
   updates the customer budget billing plan and replaces the previous amount by `new_bbp` in euros
2. `update_customer_contact_info(email_adress : optional(str), phone_number : optional(str)`
   updates customer contact information with the provided email address and/or phone number

Example Output

Your generated output should be a validation UI for this agent configuration. For this exercise, an example expectable output is illustrated below[1].

[1] This UI has been taken and slightly adapted from a Wonka AI project.



Expected Deliverables

- **Architecture diagram**: major components, responsibilities, and interactions
- **Data flow**: how agent setup data is transformed into a validation UI
- **Implementation**: a runnable prototype or minimal application that demonstrates the generation workflow
- **Technical considerations**: key design decisions and trade-offs (for example extensibility, reliability, security, observability, and performance)

Any additional depth beyond the deliverables above is considered bonus.

Constraints and Assumptions

- **Target effort**: approximately 4 hours
- **Practicality**: favor a simple but working end-to-end implementation over an over-engineered design
- **Assumptions**: explicitly state assumptions where requirements are unspecified

Reference Material

An anonymized real project document containing technical documentation is provided as supporting material in `reference.pdf`. It describes a project that follows the same 3-step framework introduced in the context section: triggering, AI agent automation, potentially involving multiple agents, and human validation UI.

API Keys

The following API keys are provided for use in your implementation. You may use either or both providers.

Anthropic

```
sk-ant-api03-MgdljxeQcKxwTPD9nxQdfA8e6hxQzbV6goQS-MMO0doiv4dQmy2ybSo3cBRUvKS
↪ mtNN1T_0vfUCZMVM3qjadWg-XNlrXgAA
```

OpenAI

```
sk-proj--SawcXAc7_gpYyTXKTwaOHwLxGNayAVZhZj_A7PT5B5LfHU0vmWZHLjMrI4HzgGD7LdR
↪ elT3kZT3BlbkFJN7jzNXSpuG0wyIqVrr5Fld1yZ7QdvCdQLaEtynbf2Gy4w5uveP8HYLJ008
↪ 5CdyZYYWYr-QPNgA
```

Submission

Provide the necessary material to review both the design and the implementation, including:

- the architecture diagram
- the data flow explanation
- the implementation source code
- the generated UI screenshot image for an example scenario

Optional Bonus

If time permits, you may include **any additional work you consider relevant** to strengthen your proposal.